

A Positive Contraction Which Is Not ℓ^1 -Contractive

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Abstract

Le Merdy and Zadeh ask whether every positive contraction between noncommutative L^p -spaces is ℓ^1 -contractive. We give a negative answer for $p = 2$ on $M_4(\mathbb{C})$. A four-dimensional map is a positive Hilbert–Schmidt contraction, whereas an explicit three-term sequence has ℓ^1 -amplification ratio $\sqrt{2/\sqrt{3}} > 1$. Every norm in the argument is computed exactly.

1 The source question

For a finite sequence (z_n) in S_4^2 , the norm used in the source is

$$\|(z_n)\|_{S_4^2(\ell^1)} = \inf_{z_n = a_n b_n} \left(\left\| \sum_n a_n a_n^* \right\|_2 \left\| \sum_n b_n^* b_n \right\|_2 \right)^{1/2}. \quad (1)$$

Here $\|\cdot\|_2$ is the Hilbert–Schmidt norm. This is the finite version of the factorization formula in Lemma 2.3 of the published paper (Lemma 2.6 in the arXiv source) [1]. Immediately after Remark 5.2, the authors write: “We do not know if any positive contraction is ℓ^1 -contractive.”

$$\leq \|T\| \left\| \sum_{n=1}^{\infty} a_n a_n^* \right\|_p^{\frac{1}{2}} \left\| \sum_{n=1}^{\infty} b_n^* b_n \right\|_p^{\frac{1}{2}} < 1.$$

This shows that T is ℓ^1 -contractive. □

Remark 5.2.

(a) An obvious consequence of Proposition 5.1 is that if T is a completely positive contraction then it is ℓ^1 -contractive.

(b) Let $\mathcal{N}^{op}$ be the opposite von Neumann algebra of $\mathcal{N}$ and let $I_{op}: L^p(\mathcal{N}) \rightarrow L^p(\mathcal{N}^{op})$ denote the identity map. We say that $T: L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N})$ is 2-copositive if the operator $I_{op} \circ T: L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N}^{op})$ is 2-positive. It is easy to check that

$$L^p(\mathcal{N}; \ell^1) = L^p(\mathcal{N}^{op}, \ell^1) \quad \text{isometrically.}$$

Therefore, Proposition 5.1 implies that any contractive 2-copositive map $L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N})$ is ℓ^1 -contractive. It therefore follows that if a positive map $L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N})$ can be written as a convex combination of a contractive 2-positive map and a contractive 2-copositive map, then it is ℓ^1 -contractive.

We do not know if any positive contraction is ℓ^1 -contractive, however we show below that positive operators are ℓ^1 -bounded.

Proposition 5.3. *Let $T: L^p(\mathcal{M}) \rightarrow L^p(\mathcal{N})$ be a bounded operator. If T is positive, then T is ℓ^1 -bounded, with $\|T\|_{\ell^1} < 4\|T\|$.*

2 Proof intuition

The skew unitary U sends each vector to an orthogonal partner. Hence, on a rank-one positive input, T subtracts two orthogonal rank-one projections from the scalar matrix that dominates both; this is the mechanism behind positivity. On the Hilbert–Schmidt space, the transpose-twist is an involutive isometry, so its orthogonal eigenspace decomposition makes the contraction estimate immediate.

The failure of ℓ^1 -contractivity is driven by row-column incidence. The three input matrices admit an efficient weighted factorization, while their images form a 2-regular bipartite pattern. Entrywise Cauchy–Schwarz gives sharp lower bounds for the two factorization norms, and the forced output norm is strictly larger than the input norm.

3 The positive contraction

Let

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} J & 0 \\ 0 & J \end{pmatrix},$$

and define $T : M_4(\mathbb{C}) \rightarrow M_4(\mathbb{C})$ by

$$T(x) = \frac{1}{2}(\text{Tr}(x)I_4 - x - Ux^\top U^*). \quad (2)$$

Lemma 1. *T is positive and $\|T\|_{S_4^2 \rightarrow S_4^2} = 1$.*

Proof. It suffices for positivity to take $x = \xi\xi^* \geq 0$. Put $\eta = U\bar{\xi}$. Since U is unitary and $U^\top = -U$,

$$\|\eta\| = \|\xi\|, \quad \xi^*\eta = \bar{\xi}^\top U\bar{\xi} = 0.$$

Consequently

$$2T(\xi\xi^*) = \|\xi\|^2 I_4 - \xi\xi^* - \eta\eta^* \geq 0.$$

Every positive matrix is a sum of positive rank-one matrices, so T is positive.

For the norm, set $\mathcal{R}(x) = Ux^\top U^*$ and $P(x) = \text{Tr}(x)I_4/4$. The map $\mathcal{R}$ is a Hilbert–Schmidt unitary involution, hence a self-adjoint involution, and it fixes I_4 . Thus

$$T = 2P - \frac{I + \mathcal{R}}{2}$$

acts on the mutually orthogonal spaces

$$\mathbb{C}I_4, \quad \{x : \text{Tr } x = 0, \mathcal{R}x = x\}, \quad \{x : \mathcal{R}x = -x\}$$

by the scalars 1, -1 , 0 , respectively. Hence $\|T\| = 1$. □

4 The exact amplification ratio

Write E_{ij} for the standard matrix units and set

$$x_1 = E_{41} - E_{23}, \quad x_2 = -(E_{11} + E_{22}), \quad x_3 = E_{31} + E_{24}. \quad (3)$$

Directly from $Ue_1 = -e_2$, $Ue_2 = e_1$, $Ue_3 = -e_4$ and $Ue_4 = e_3$, one obtains

$$y_1 := T(x_1) = -x_1, \quad y_2 := T(x_2) = -(E_{33} + E_{44}), \quad y_3 := T(x_3) = -x_3. \quad (4)$$

Lemma 2.

$$\|(x_1, x_2, x_3)\|_{S_4^2(\ell^1)} = \sqrt{6\sqrt{3}}.$$

Proof. For the upper bound, let

$$\begin{aligned} a_1 &= 3^{1/4}E_{41} - E_{23}, & b_1 &= 3^{-1/4}E_{11} + E_{33}, \\ a_2 &= -(3^{1/4}E_{11} + E_{22}), & b_2 &= 3^{-1/4}E_{11} + E_{22}, \\ a_3 &= 3^{1/4}E_{31} + E_{24}, & b_3 &= 3^{-1/4}E_{11} + E_{44}. \end{aligned}$$

Then $x_n = a_n b_n$ and

$$\sum a_n a_n^* = \text{diag}(\sqrt{3}, 3, \sqrt{3}, \sqrt{3}), \quad \sum b_n^* b_n = \text{diag}(\sqrt{3}, 1, 1, 1).$$

Their Hilbert–Schmidt norms are $\sqrt{18}$ and $\sqrt{6}$, giving the claimed upper bound through (1).

For the reverse inequality, consider an arbitrary factorization $x_n = c_n d_n$, and put

$$\alpha_{n,i} = (c_n c_n^*)_{ii}, \quad \beta_{n,j} = (d_n^* d_n)_{jj}.$$

Entrywise Cauchy–Schwarz gives $|(x_n)_{ij}|^2 \leq \alpha_{n,i} \beta_{n,j}$. For the six nonzero entries in (3), define

$$\begin{aligned} (p_1, p_2, p_3) &= (\alpha_{1,4}, \alpha_{2,1}, \alpha_{3,3}), & (q_1, q_2, q_3) &= (\alpha_{1,2}, \alpha_{2,2}, \alpha_{3,2}), \\ (r_1, r_2, r_3) &= (\beta_{1,1}, \beta_{2,1}, \beta_{3,1}), & (s_1, s_2, s_3) &= (\beta_{1,3}, \beta_{2,2}, \beta_{3,4}). \end{aligned}$$

Then $p_n r_n \geq 1$ and $q_n s_n \geq 1$. If

$$P = (\sum p_n^2)^{1/2}, \quad Q = \sum q_n, \quad R = \sum r_n, \quad S = (\sum s_n^2)^{1/2},$$

the diagonal entries imply

$$\left\| \sum c_n c_n^* \right\|_2 \geq \sqrt{P^2 + Q^2}, \quad \left\| \sum d_n^* d_n \right\|_2 \geq \sqrt{R^2 + S^2}.$$

Moreover, Cauchy–Schwarz gives

$$PR \geq \left(\sum p_n^2 \right)^{1/2} \sum \frac{1}{p_n} \geq 3\sqrt{3}, \quad QS \geq \sum q_n \left(\sum \frac{1}{q_n^2} \right)^{1/2} \geq 3\sqrt{3}.$$

Therefore the product of the last two Hilbert–Schmidt norms is at least

$$\sqrt{P^2 + Q^2} \sqrt{R^2 + S^2} \geq PR + QS \geq 6\sqrt{3}.$$

Taking the square root and then the infimum in (1) proves the lower bound. $\square$

Lemma 3.

$$\|(y_1, y_2, y_3)\|_{S_4^2(\ell^1)} = 2\sqrt{3}.$$

Proof. For the upper bound factor $y_n = c_n d_n$ with

$$(c_1, c_2, c_3) = (y_1, y_2, y_3), \quad (d_1, d_2, d_3) = (E_{11} + E_{33}, E_{33} + E_{44}, E_{11} + E_{44}).$$

Then

$$\sum c_n c_n^* = \text{diag}(0, 2, 2, 2), \quad \sum d_n^* d_n = \text{diag}(2, 0, 2, 2),$$

and both Hilbert–Schmidt norms equal $\sqrt{12}$.

For the lower bound, take any factorization and attach to each of the six unit-modulus entries of (y_1, y_2, y_3) the quantities $\alpha_e = (c_n c_n^*)_{ii}$ and $\beta_e = (d_n^* d_n)_{jj}$ at its row and column. Thus $\alpha_e \beta_e \geq 1$. The six entries form a 2-regular bipartite graph on the three rows $\{2, 3, 4\}$ and three columns $\{1, 3, 4\}$. Let u_i and v_j be the sums of the incident α_e and β_e , respectively. Then

$$6 \leq \sum_e \sqrt{\alpha_e \beta_e} \leq \sqrt{\left(\sum_i u_i\right) \left(\sum_j v_j\right)}.$$

Since $\sum_i u_i \leq \sqrt{3} \|u\|_2$ and similarly for v , while the diagonal entries give

$$\left\| \sum c_n c_n^* \right\|_2 \geq \|u\|_2, \quad \left\| \sum d_n^* d_n \right\|_2 \geq \|v\|_2,$$

the product of these Hilbert–Schmidt norms is at least 12. Formula (1) proves the result. $\square$

Theorem 1. *There is a positive contraction on a noncommutative L^2 -space which is not ℓ^1 -contractive. In particular, the question of Le Merdy and Zadeh has a negative answer.*

Proof. Use T from (2). Lemma 1 shows that it is a positive contraction on S_4^2 . Lemmas 2 and 3 give

$$\frac{\|(T(x_1), T(x_2), T(x_3))\|_{S_4^2(\ell^1)}}{\|(x_1, x_2, x_3)\|_{S_4^2(\ell^1)}} = \frac{2\sqrt{3}}{\sqrt{6\sqrt{3}}} = \sqrt{\frac{2}{\sqrt{3}}} > 1.$$

$\square$

5 Scope and review notes

The counterexample uses the unnormalized trace on $M_4(\mathbb{C})$ and $p = 2$, both allowed by the source setup. The proof is finite-dimensional and exact; the numerical search that suggested the three matrices is not used. A bounded exact-phrase and citing-work search performed on 27 August 2026 found no later resolution of the question. This is a provisional novelty check, not a definitive literature claim.

References

- [1] C. Le Merdy and S. Zadeh, ℓ^1 -contractive maps on noncommutative L^p -spaces, J. Operator Theory **85** (2021), 417–442; arXiv:1907.03995.